

- ☐ Preliminary Specifications
☒ Final Specifications

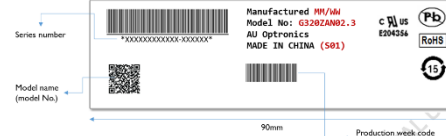
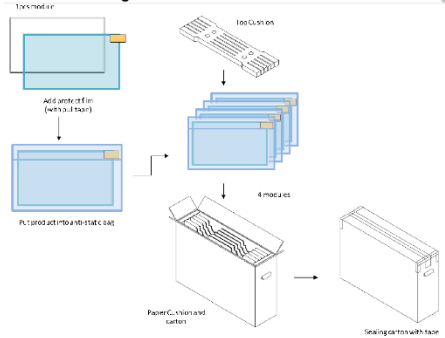
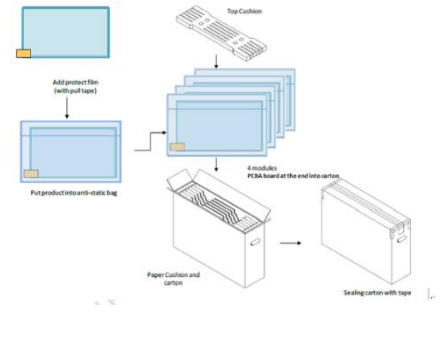
Module	32 Inch Color TFT-LCD
Model Name	G320ZAN02.3

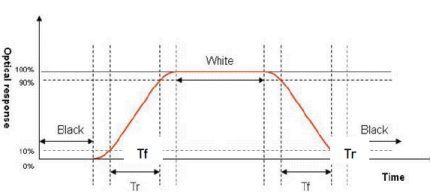
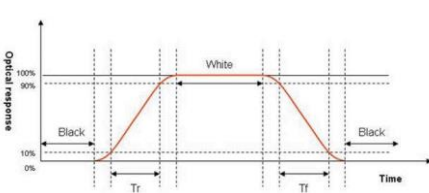
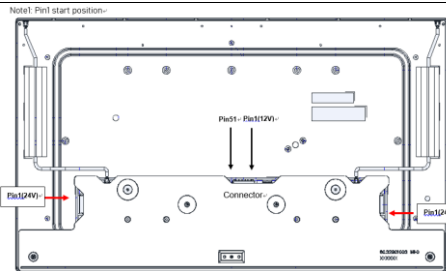
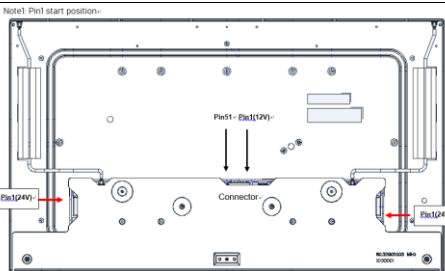
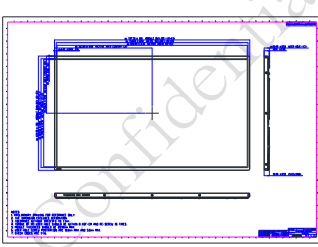
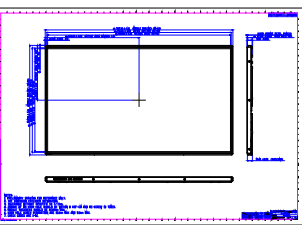
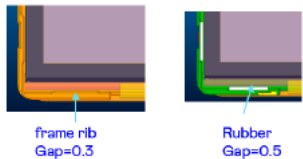
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General Display Business Unit / AUO Display Plus corporation																	

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Record of Revision

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		All		Final spec																																																																																																																												
		P27	9.1 Shipping Label (on the rear side of TFT-LCD display) 	9.1 Shipping Label (on the rear side of TFT-LCD display). 																																																																																																																												
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		P9		
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		P30	W2.2 Materials: The materials used in the product shall be of the highest quality and shall be suitable for use in the product. W2.3 Assembly: The product shall be assembled in accordance with the requirements of the specification. W2.4 Inspection: The product shall be inspected in accordance with the requirements of the specification. W2.5 Packaging: The product shall be packaged in accordance with the requirements of the specification.	W2.1 Sharp Edge Requirements: There shall be no sharp edges or corners on the product which could cause injury. W2.2 Materials: The materials used in the product shall be of the highest quality and shall be suitable for use in the product. W2.3 Assembly: The product shall be assembled in accordance with the requirements of the specification. W2.4 Inspection: The product shall be inspected in accordance with the requirements of the specification. W2.5 Packaging: The product shall be packaged in accordance with the requirements of the specification.
		All		
I.0	9/25	P06	727.3 (H) x 424.5 (V) x 20.48 (D) (Typ)	727.3 (H) x 424.5 (V) x 21.37 (D) (Typ)
		P26		
		NA	NA	

I. Operating Precautions

- 1) Since front polarizer is easily damaged, please be cautious and not to scratch it.
- 2) Be sure to turn off power supply when inserting or disconnecting from input connector.
- 3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
- 4) When the panel surface is soiled, wipe it with absorbent cotton or soft cloth.
- 5) Since the panel is made of glass, it may be broken or cracked if dropped or bumped on hard surface.
- 6) To avoid ESD (Electro Static Discharge) damage, be sure to ground yourself before handling TFT-LCD Module.
- 7) Do not open nor modify the module assembly.
- 8) Do not press the reflector sheet at the back of the module to any direction.
- 9) In case if a module has to be put back into the packing container slot after it was taken out from the container, do not press the center of the LED light bar edge. Instead, press at the far ends of the LED light bar edge softly. Otherwise the TFT Module may be damaged.
- 10) At the insertion or removal of the Signal Interface Connector, be sure not to rotate nor tilt the Interface Connector of the TFT Module.
- 11) TFT-LCD Module is not allowed to be twisted & bent even force is added on module in a very short time. Please design your display product well to avoid external force applying to module by end-user directly.
- 12) Small amount of materials having no flammability grade is used in the LCD module. The LCD module should be supplied by power complied with requirements of Limited Power Source (IEC60950-1 or UL1950), or be applied exemption.
- 13) Severe temperature condition may result in different luminance, response time and lamp ignition voltage.
- 14) Continuous operating TFT-LCD display under low temperature environment may accelerate lamp exhaustion and reduce luminance dramatically.
- 15) The data on this specification sheet is applicable when LCD module is placed in landscape position.
- 16) Continuous displaying fixed pattern may induce image sticking. It's recommended to use screen saver or shuffle content periodically if fixed pattern is displayed on the screen.

2. General Description

This specification applies to the Color Active Matrix Liquid Crystal Display G320ZAN02.3 composed of a TFT-LCD display, a driver and power supply circuit, and a LED backlight system. The screen format is intended to support the UHD (3840(H)x2160(V)) screen and 1.07B colors. All input signals are eDP interface compatible. LED driving board for backlight unit is included in G320ZAN02.3.

2.1 Display Characteristics

The following items are characteristics summary on the table under 25°C condition:

Items	Unit	Specifications
Screen Diagonal	[inch]	32.0" (812.8mm)
Active Area	[mm]	708.48 (H) x 398.52 (V)
Resolution		3840(x3) x 2160
Pixel Pitch	[mm]	0.1845 (per one triad) x 0.1845
Pixel Arrangement		R.G.B. Vertical Stripe
Display Mode		Normally Black, AHVA
Nominal Input Voltage VDD	[Volt]	+12.0 V
Power Consumption	[Watt]	Max.133W (Logi.c: max. 32W + BL power: max.101W)
Weight	[Grams]	4630 (Typ)
Physical Size	[mm]	727.3 (H) x 424.5 (V) x 21.37 (D) (Typ)
Electrical Interface		eDP(4lane)
Surface Treatment		Glare treatment
Support Color		10bit(True 10)
Temperature Range		
Operating	[°C]	0 to +50
Storage (Non-Operating)	[°C]	-20 to +60
RoHS Compliance		Yes

2.2 Optical Characteristics

The optical characteristics are measured under stable conditions(warm up 30 mins) at 25°C(Room Temperature), VDD 12V , Frame rate:60Hz:

Item	Unit	Conditions		Min.	Typ.	Max.	Note
White Luminance	cd/m ²	ILED=95mA(center point)		680	850	---	1
Uniformity	%	9 points		---	---	20	2,3
Contrast Ratio	--			1050	1500		4
Response Time	msec	Rising		---	10	20	5
		Falling		---	10	20	
		Rising + Falling		---	20	40	
Viewing Angle	degree	Horizontal CR >= 10	(Right)	75	89	---	6
			(Left)	75	89	---	
		Vertical CR >= 10	(Upper)	75	89	---	
			(Lower)	75	89	---	
Color / Chromaticity Coordinates (CIE 1931)	--	Red x		0.657	0.687	0.717	
		Red y		0.278	0.308	0.338	
		Green x		0.181	0.211	0.241	
		Green y		0.700	0.730	0.760	
		Blue x		0.119	0.149	0.179	
		Blue y		0.021	0.051	0.081	
		White x		0.292	0.322	0.352	
		White y		0.294	0.324	0.354	
Adobe RGB coverage ratio	%(typ)			DCI : 98% / Adobe RGB : 100%			

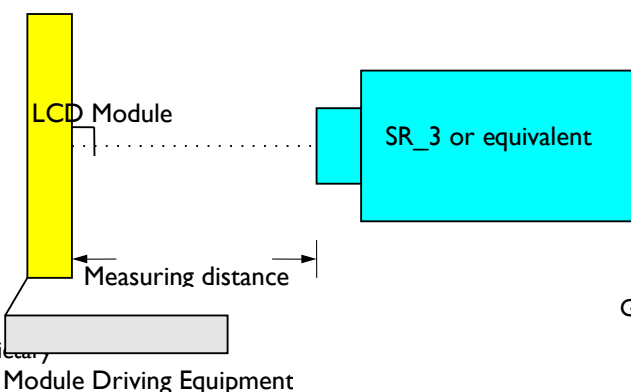
Note 1: Measurement method

Equipment Pattern Generator, Power Supply, Digital Voltmeter, Luminance meter (SR_3 or equivalent)

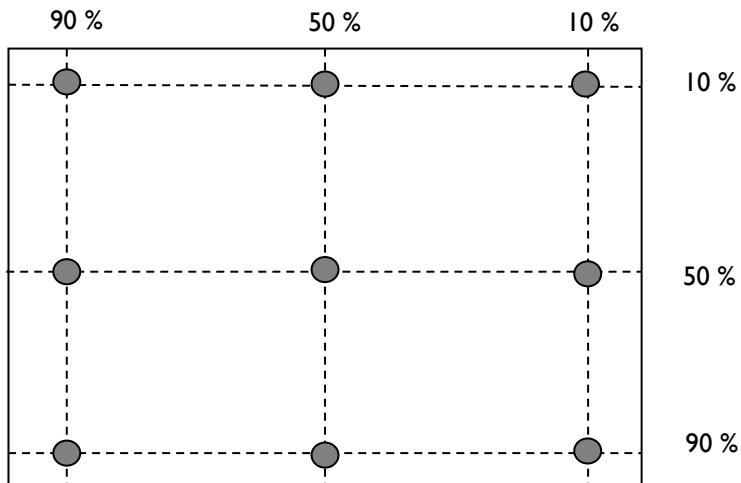
Aperture 10 with 50cm viewing distance

Test Point Center

Environment < 1 lux



Note 2: Definition of 9 points position



Note 3: Definition of luminance uniformity of 9 points.

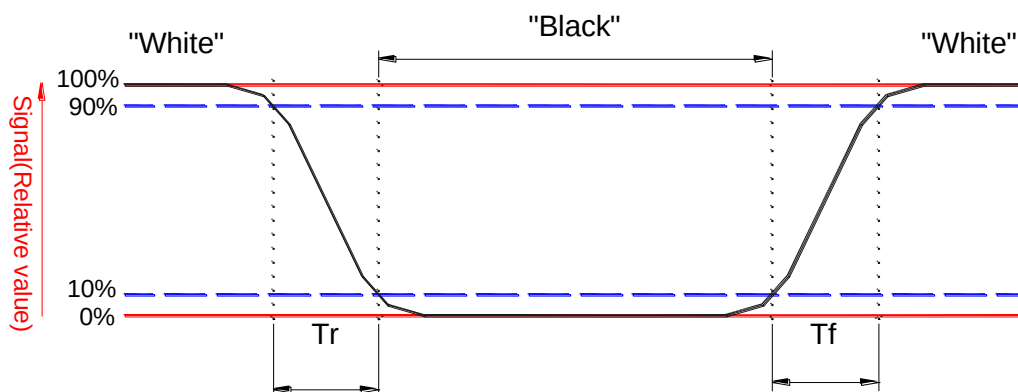
$$\delta_{w9} = \frac{|B_{\max} \text{ or } B_{\min} - B_{\text{avg}}|}{B_{\text{avg}}} \times 100\%$$

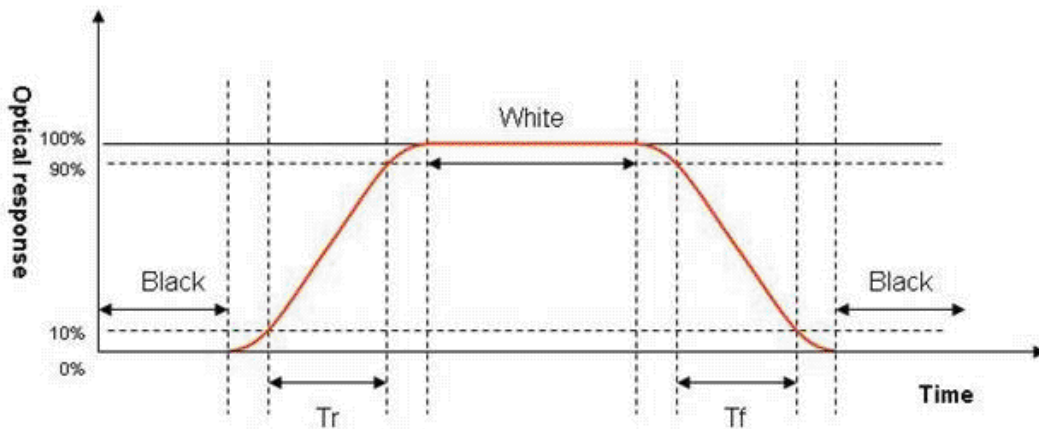
Note 4: Definition of contrast ratio (CR):

$$\text{Contrast ratio (CR)} = \frac{\text{Brightness on the "White" state}}{\text{Brightness on the "Black" state}}$$

Note 5: Definition of response time:

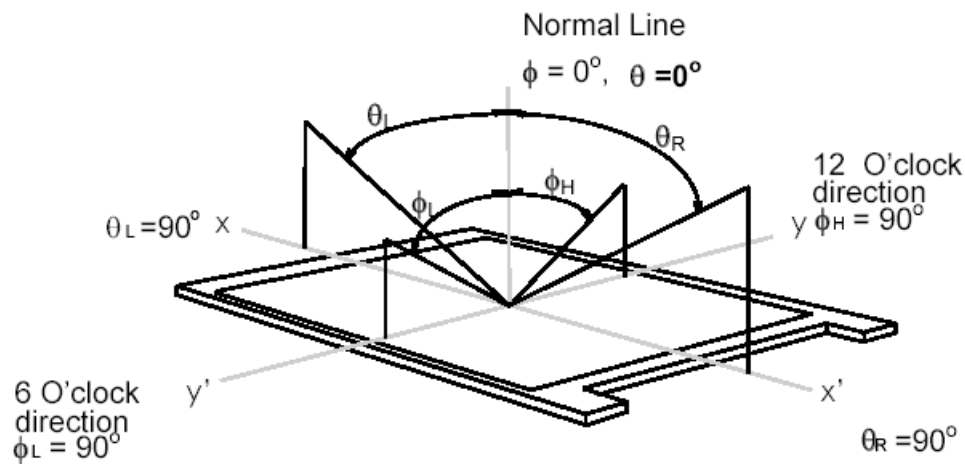
The output signals of photo detector are measured when the input signals are changed from "White" to "Black" (falling time) and from "Black" to "White" (rising time), respectively. The response time interval is between 10% and 90% of amplitudes. Please refer to the figure as below.





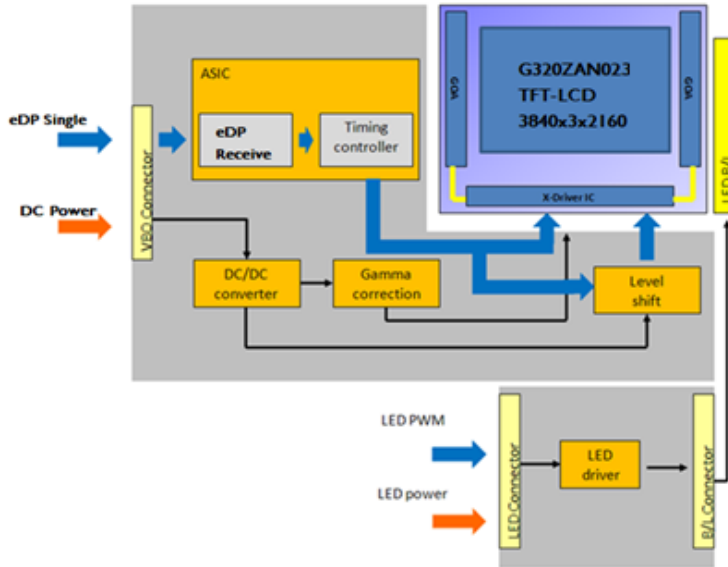
Note 6: Definition of viewing angle

Viewing angle is the measurement of contrast ratio ≥ 10 , at the screen center, over a 180° horizontal and 180° vertical range (off-normal viewing angles). The 180° viewing angle range is broken down as below: 90° (θ) horizontal left and right, and 90° (ϕ) vertical high (up) and low (down). The measurement direction is typically perpendicular to the display surface with the screen rotated to its center to develop the desired measurement viewing angle.



3. Functional Block Diagram

The following diagram shows the functional block of the 32 inch color TFT/LCD module:



3.1 Interface Connection :

Connector Name / Designation	Signal Connector
Manufacturer	IAE
Connector Model Number	FI-RE51S-HF
Adaptable Plug	FI-RE51CL

4. Absolute Maximum Ratings

4.1 Absolute Ratings of TFT LCD Module

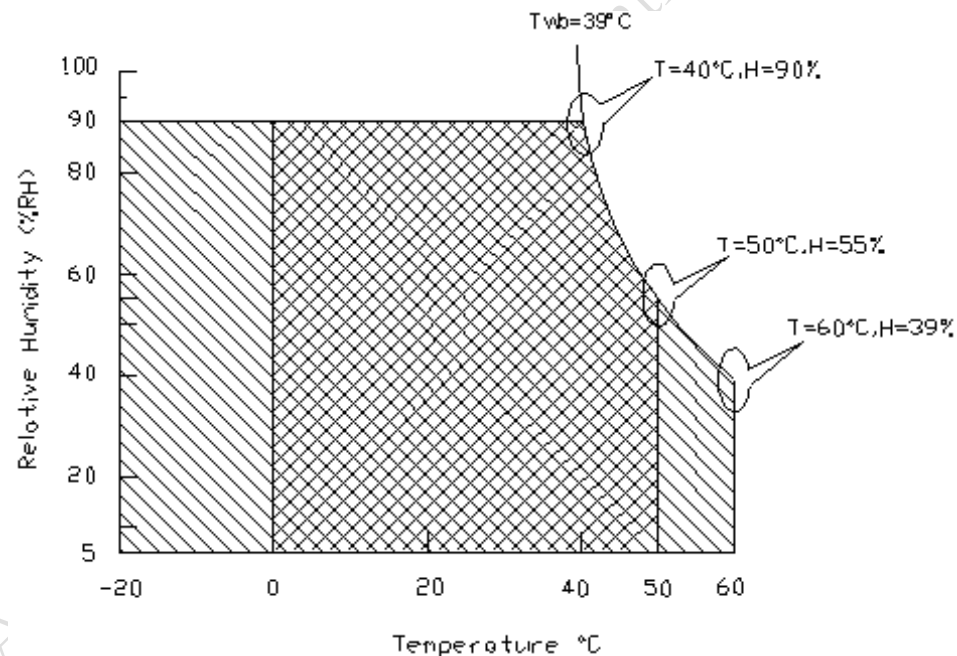
Item	Symbol	Min	Max	Unit
Logic/LCD drive Voltage	Vin	-0.3	+13.5	[Volt]

4.2 Absolute Ratings of Environment

Item	Symbol	Min	Max	Unit
Operating Temperature	TOP	0	+50	[°C]
Operation Humidity	HOP	5	90	[%RH]
Storage Temperature	TST	-20	+60	[°C]
Storage Humidity	HST	5	90	[%RH]
Panel Surface Temperature	PST		60	[°C]

Note 1: Maximum Wet-Bulb should be 39°C and no condensation.

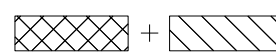
Note 2: Surface temperature is measured at 50°C, dry condition.



Operating Range



Storage Range



5. Electrical Characteristics

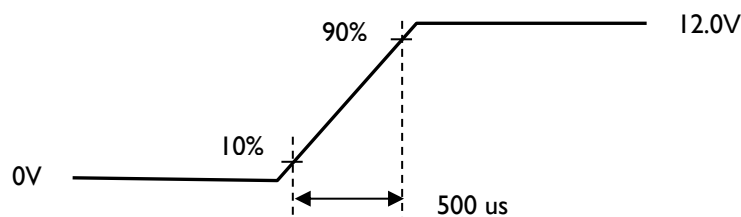
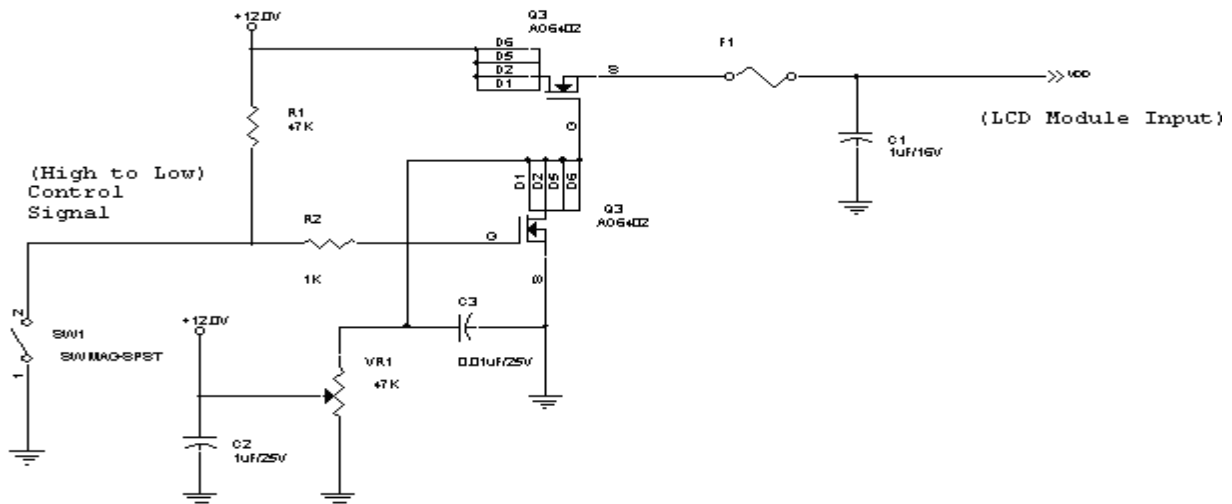
5.1 TFT LCD Module

5.1.1 Power Specification

Input power specifications are shown as follows;

Symbol	Parameter	Min	Typ	Max	Units	Remark
VDD	Logic/LCD Drive Voltage	10.8	12.0	13.2	[Volt]	±10%
IDD	VDD Current	-	2	2.4	[A]	White Pattern (VDD=12V, at 60Hz)
Irush	LCD Inrush Current	-	-	4	[A]	Note 1
PDD	VDD Power	-	24	32	[Watt]	White Pattern (VDD=12V, at 60Hz)
VDDrp	Allowable Logic/LCD Drive Ripple Voltage	-	-	VDD* 5%	[mV]	VDD= 12.0V, White pattern, Fv=60Hz

Note 1: Measurement condition:



VDD rising time

5.2 Backlight Unit

5.2.1 LED Backlight Unit: Driver Connector

Connector Name / Designation	Lamp Connector
Manufacturer	JST
Connector Model Number	SI4B-PH-SM6-K-TB(HF)
Mating Model Number	PHR-14

Pin #	Symbol	Pin Description
1	VBL	+24V
2	VBL	+24V
3	VBL	+24V
4	VBL	+24V
5	VBL	+24V
6	GND	GND
7	GND	GND
8	GND	GND
9	GND	GND
10	GND	GND
11	BL_DIM_A	Analog Dimming
12	BL_EN	BL enable (3.3V-On / 0V-Off)
13	NC	N/A
14	BL_DIM_P	PWM Dimming

Note 1: Use the PWM mode. The pin.11 must be NC.

Note 2: Use the Analog mode. The pin.14 must be NC.

5.2.2 Parameter guideline for LED

Following characteristics are measured under a stable condition using an inverter at 25°C(Room Temperature):

LED characteristics

Symbol	Parameter	Min	Typ	Max	Units	Condition
PLED	Backlight Power Consumption	---	72.96	82.69	[Watt]	LED only
LTLED	LED Life-Time	30,000	---	---	Hour	LED only

Note 1: Calculator value for reference $P_{LED} = V_F$ (Normal Distribution) * I_F (Normal Distribution)

Note 2: The LED life-time define as the estimated time to 50% degradation of initial luminous.

Backlight input signal characteristics

Symbol	Parameter	Min.	Typ.	Max.	Unit	Remark
VDD	Input Voltage	21.6	24.0	26.4	[Volt]	
I _{VDD}	Input Current	- -	3.2	3.84	[A]	100% PWM Duty
P _{VDD}	Power Consumption	---	76.8	101.4	[Watt]	100% PWM Duty
Backlight on/off	On control Voltage	2	3.3	5	[Volt]	
	Off control Voltage		---	0.8	[Volt]	
Backlight Dimming (Analog Mode)	Dimming Voltage	0	---	3	[Volt]	Adjustable Dimming Range
Backlight Dimming (PWM Mode)	Dimming Frequency	0.2		15	[kHz]	
	Swing Voltage	3	---	3.6	V	Note3
	Dimming Duty Cycle	10	-	100	%	
I _F (one channel)	LED Forward Current	---	95	---	mA	Ta = 25°C

Note 1: Ta means ambient temperature of TFT-LCD module.

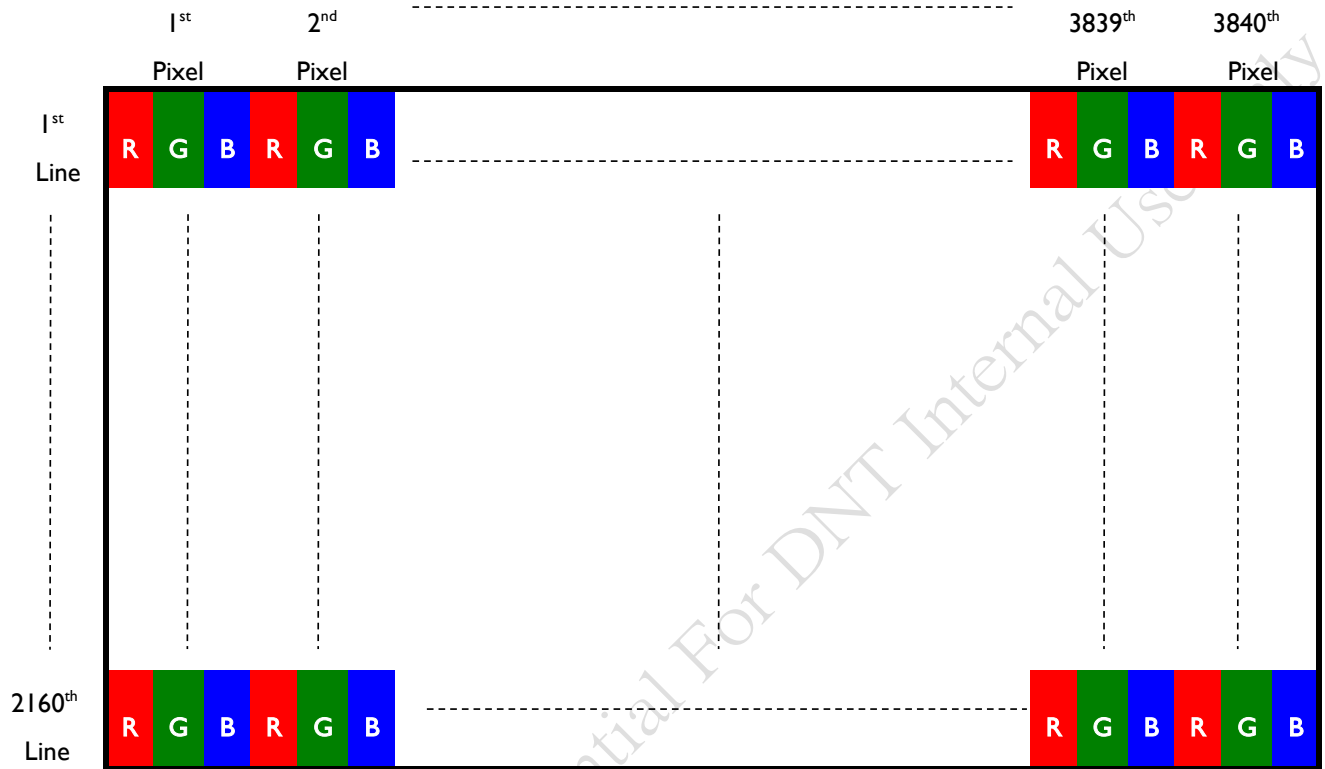
Note 2: VDD, IVDD, PVDD , I_{rush} LED are defined for LED B/L.(100% duty of PWM dimming).

Note 3: Swing voltage 3V is best linear of PWM mode.

6. Signal Characteristic

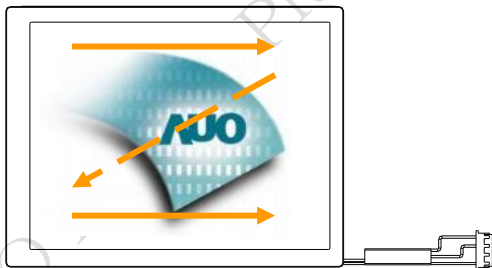
6.1 Pixel Format Image

Following figure shows the relationship between input signal and LCD pixel format.



6.2 Scanning Direction

The following figures show the image seen from the front view. The arrow indicates the direction of scan.



6.3 Signal Description

The module uses a eDP receiver embedded in AUO's ASIC. eDP is a differential signal technology for LCD interface and a high-speed data transfer device.

6.3.1 TFT LCD Module: LCD Connector

Connector Name / Designation	Signal Connector
Manufacturer	IAE
Connector Model Number	FI-RE51S-HF
Adaptable Plug	FI-RE51CL

PIN #	Symbol	Description
1	VDD	Power +12V
2	VDD	Power +12V
3	VDD	Power +12V
4	VDD	Power +12V
5	VDD	Power +12V
6	GND	Ground
7	GND	Ground
8	GND	Ground
9	NC	No connection (for AUO test only. Do not connect)
10	NC	No connection (for AUO test only. Do not connect)
11	NC	No connection (for AUO test only. Do not connect)
12	NC	No connection (for AUO test only. Do not connect)
13	NC	No connection (for AUO test only. Do not connect)
14	NC	No connection (for AUO test only. Do not connect)
15	NC	No connection (for AUO test only. Do not connect)
16	NC	No connection (for AUO test only. Do not connect)
17	GND	Ground
18	Ist Lane3_N	Negative eDP differential data input
19	Ist Lane3_P	Positive eDP differential data input
20	GND	Ground
21	Ist Lane2_N	Negative eDP differential data input
22	Ist Lane2_P	Positive eDP differential data input
23	GND	Ground
24	Ist Lane1_N	Negative eDP differential data input
25	Ist Lane1_P	Positive eDP differential data input
26	GND	Ground
27	Ist Lane0_N	Negative eDP differential data input
28	Ist Lane0_P	Positive eDP differential data input

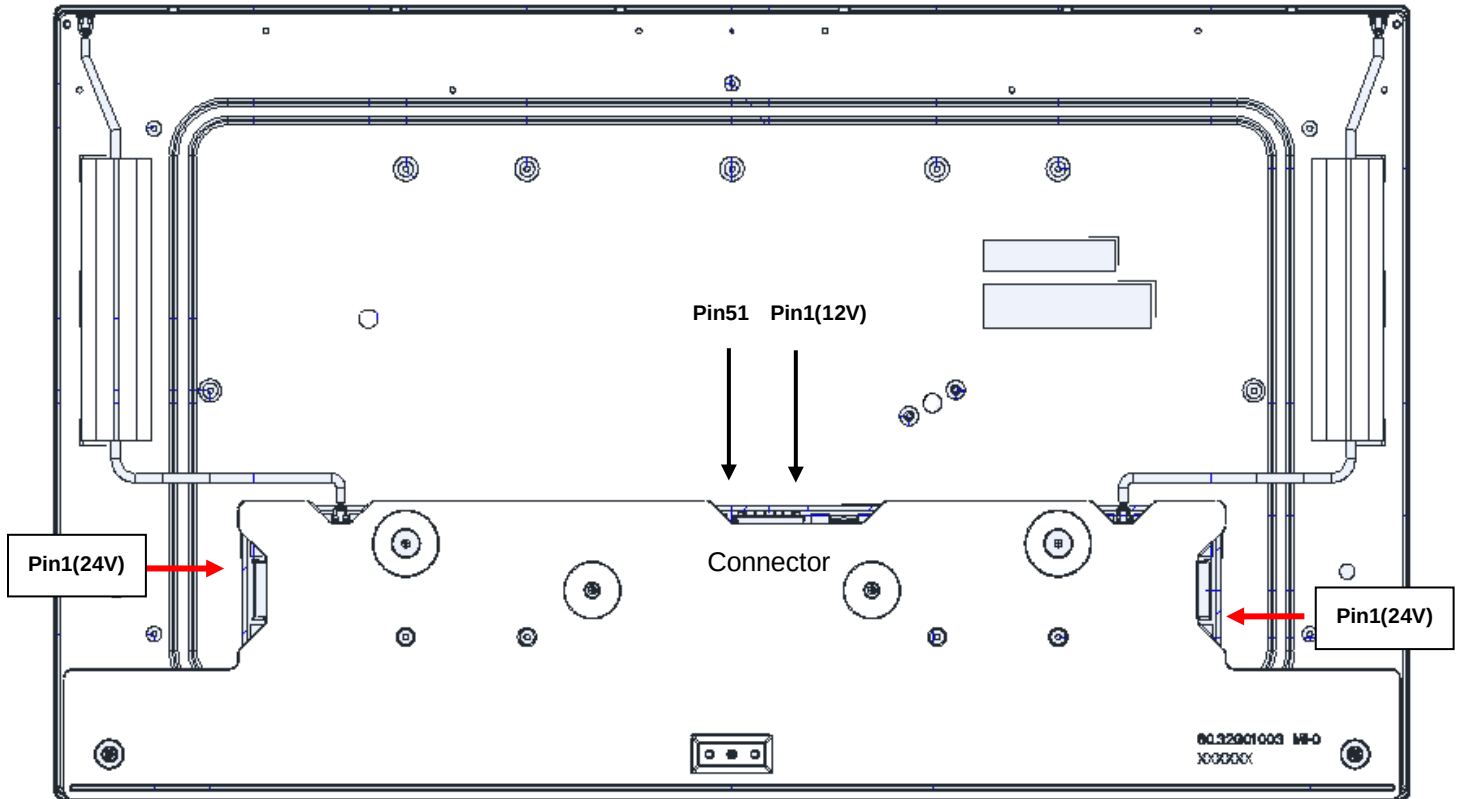
29	GND	Ground
30	Ist AUX_CH_P	Positive AUX Channel differential data input
31	Ist AUX_CH_N	Negative AUX Channel differential data input
32	GND	Ground
33	NC	No connection (for AUO test only. Do not connect)
34	GND	Ground
35	NC	No connection (for AUO test only. Do not connect)
36	NC	No connection (for AUO test only. Do not connect)
37	GND	Ground
38	NC	No connection (for AUO test only. Do not connect)
39	NC	No connection (for AUO test only. Do not connect)
40	GND	Ground
41	NC	No connection (for AUO test only. Do not connect)
42	NC	No connection (for AUO test only. Do not connect)
43	GND	Ground
44	NC	No connection (for AUO test only. Do not connect)
45	NC	No connection (for AUO test only. Do not connect)
46	GND	Ground
47	NC	No connection (for AUO test only. Do not connect)
48	NC	No connection (for AUO test only. Do not connect)
49	GND	Ground
50	HPD	Hot plug detection
51	GND	Ground

Note 1: Input Signals shall be in low status when VDD is off.

Note 2: High stands for “3.3V”, Low stands for “0V”, NC means “No Connection”.

Note 3: RSV means “Reserved”.

Note I: Pin I start position



6.4 The Input Data Format

6.4.1 eDP Data Format

The brightness of each primary color is based on the 10bit gray scale data input for the color; the higher the binary input, the brighter the color. The table below provides a reference for color versus data input.

Lane 0	Lane 1	Lane 2	Lane 3
R0-9:2	R1-9:2	R2-9:2	R3-9:2
R0-1:0/G0-9:4	R1-1:0/G1-9:4	R2-1:0/G2-9:4	R3-1:0/G3-9:4
G0-3:0/B0-9:6	G1-3:0/B1-9:6	G2-3:0/B2-9:6	G3-3:0/B3-9:6
B0-5:0/R4-9:8	B1-5:0/R5-9:8	B2-5:0/R6-9:8	B3-5:0/R7-9:8
R4-7:0	R5-7:0	R6-7:0	R7-7:0
G4-9:2	G5-9:2	G6-9:2	G7-9:2
G4-1:0/B4-9:4	G5-1:0/B5-9:4	G6-1:0/B6-9:4	G7-1:0/B7-9:4
B4-3:0/R8-9:6	B5-3:0/R9-9:6	B6-3:0/R10-9:6	B7-3:0/R11-9:6
R8-5:0/G8-9:8	R9-5:0/G9-9:8	R10-5:0/G10-9:8	R11-5:0/G11-9:8
G8-7:0	G9-7:0	G10-7:0	G11-7:0
B8-9:2	B9-9:2	B10-9:2	B11-9:2
B8-1:0/R12-9:4	B9-1:0/R13-9:4	B10-1:0/R14-9:4	B11-1:0/R15-9:4
R12-3:0/G12-9:6	R13-3:0/G13-9:6	R14-3:0/G14-9:6	R15-3:0/G15-9:6
G12-5:0/B12-9:8	G13-5:0/B13-9:8	G14-5:0/B14-9:8	G15-5:0/B15-9:8
B12-7:0	B13-7:0	B14-7:0	B15-7:0

10bit RGB to a 4-Lane Main-Link Mapping

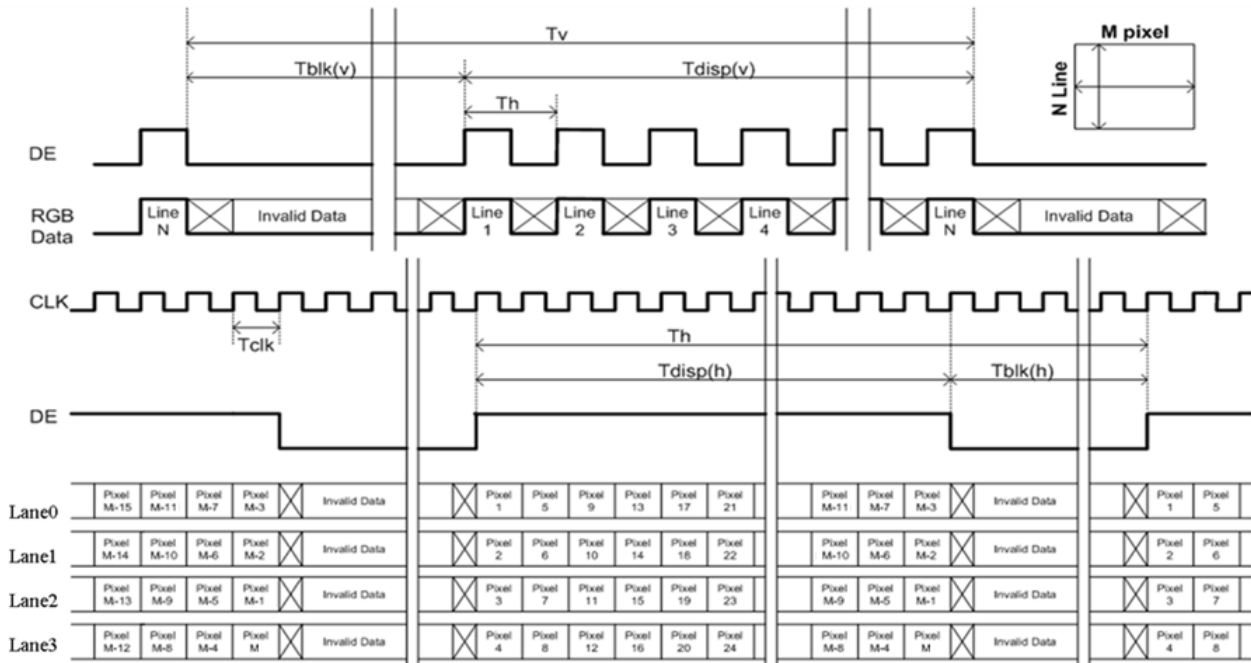
6.4.2 The Input Data Format(eDP)

The following table is for color versus input data (10bit). The higher the gray level, the brighter the color.

Color	Gary Level	Color Input Data																														Remark	
		RED data (MSB:R9,LSB:R0)										GREEN data (MSB:G9,LSB:G0)										BLUE data (MSB:B9,LSB:B0)											
		R9	R8	R7	R6	R5	R4	R3	R2	R1	R0	G9	G8	G7	G6	G5	G4	G3	G2	G1	G0	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0		
Black	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
White	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
L511	-	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	
Red	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black	
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L1023	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Green	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black	
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L1023	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	
Blue	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black	
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L1023	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	

6.4.3 Timing Diagram

(Lane0~3 eDP data:1, 2, 3, 4, ...,3837,3838,3839,3840)



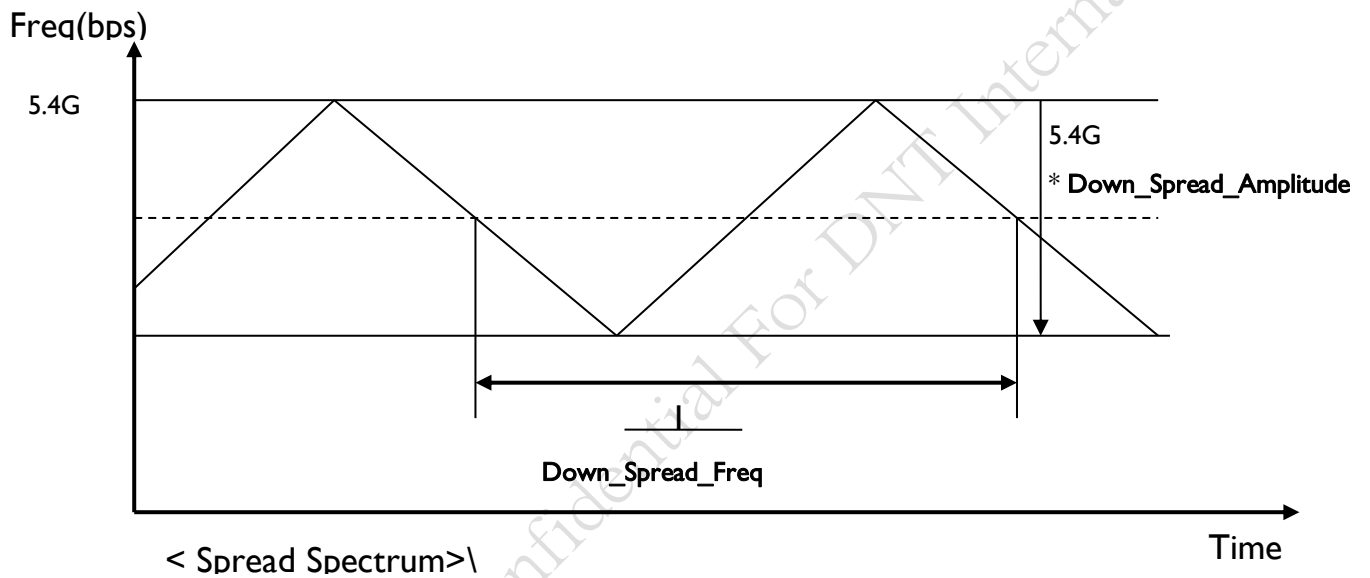
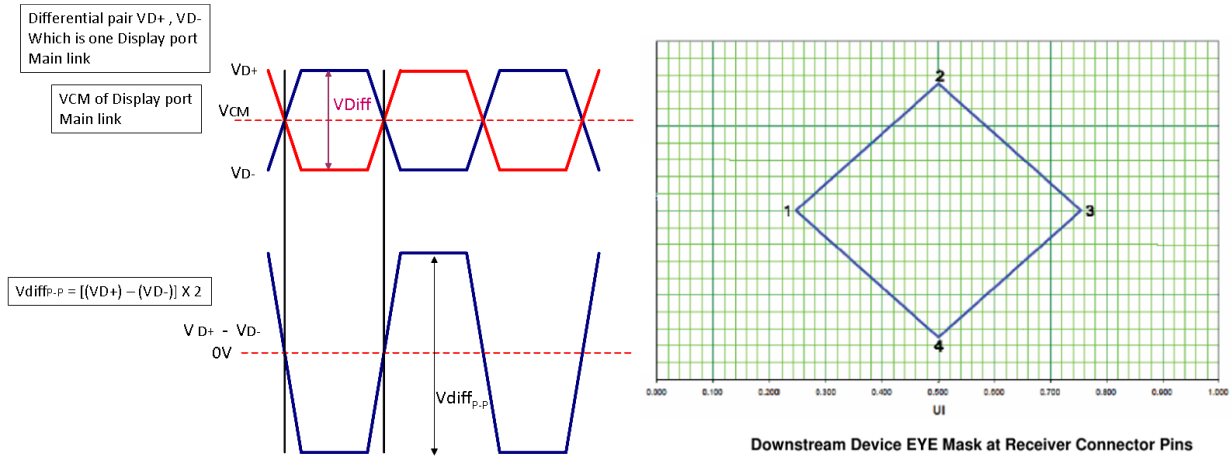
6.5 eDP Specification (Follow VESA Display Port Standard Version 1.2)

6.5.1 DisplayPort Main Link Signal:

DisplayPort Main Link		Min	Typ	Max	unit
Frequency	Main Link Frequency	-	5.4	-	Gbps
UI	Unit Interval	-	185	-	ps
VCM	RX input DC Common Mode Voltage	-	0	-	[Volt]
$V_{Diff,P-P}$	Peak-to-peak Voltage at a receiving Device	70	-	-	[mVolt]
Down_Spread_Freq	Link clock down spread frequency	30	-	33	KHz
Down_Spread_Amplitude	Link clock down spread amplitude	-	-	0.5	%

Point	Time(UI)	Voltage (V)
1	0.310	0
2	0.375~0.625	35mV
3	0.310	0
4	0.375~0.625	-35mV

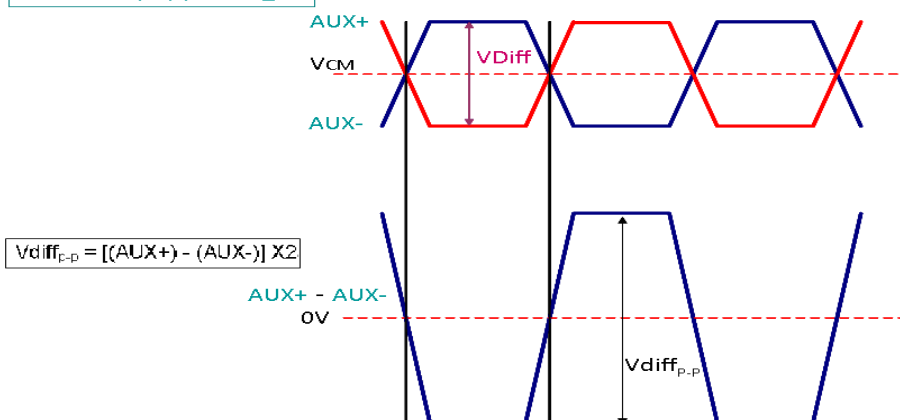
Downstream Device EYE Mask at Receiver Connector for HBR2



6.5.2 DisplayPort AUX_CH Signal:

DisplayPort AUX_CH					
		Min	Typ	Max	unit
VCM	AUX DC Common Mode Voltage	-	0	-	[Volt]
VDiff _{P-P}	AUX Peak-to-peak voltage at a receiving device	70	-	-	[mVolt]

Differential AUX+ , AUX-
Which is Display pot AUX_CH



6.5.3 DisplayPort VHPD Signal:

Display Port VHPD					
		Min	Typ	Max	unit
VHPD	HPD Voltage	2.25	-	3.6	[Volt]

6.5.4 Intra-Pair Skew:

LRX-SKEW-INTRA_PAIR					
		Min	Typ	Max	unit
LRX-SKEW-INTRA_PAIR	Lane Intra-pair Skew Tolerance	-	-	50	[ps]

6.5.5 Inter-Pair Skew

LRX-SKEW-INTER_PAIR					
		Min	Typ	Max	unit
LRX-SKEW-INTER_PAIR	Lane-to-Lane Skew at RX package pins	-	-	5700	[ps]

6.6 Interface Timing

6.6.1 Timing Characteristics

The input timing is shown as the following table.

Symbol	Description		Min.	Typ.	Max.	Unit	Remark
Tv	Vertical Section	Period	2184	2200	2280	Th	
Tdisp (v)		Active	2160	2160	2160	Th	
Tblk (v)		Blanking	24	40	120	Th	
Fv		Frequency	49	60	61	Hz	Note 6-6
Th	Horizontal Section	Period	3920	4120	4240	Tclk	
Tdisp (h)		Active	3840	3840	3840	Tclk	
Tblk (h)		Blanking	80	280	400	Tclk	
Fh		Frequency	107.1	132	144.4	kHz	Note 6-4
Tclk	Pixel Clock	Period	1.77	1.84	2.38	ns	1/Fclk
Fclk		Frequency	419.50	543.8	566.28	MHz	Note 6-5
Link Rate per Lane			5.4			Gbps	5.4

Note 6-4: The equation is listed as following. Please don't exceed the above recommended value.

$$\begin{aligned} F_h (\text{Min.}) &= F_{\text{clk}} (\text{Min.}) / T_h (\text{Min.}) \\ F_h (\text{Typ.}) &= F_{\text{clk}} (\text{Typ.}) / T_h (\text{Typ.}) \\ F_h (\text{Max.}) &= F_{\text{clk}} (\text{Max.}) / T_h (\text{Min.}) \end{aligned}$$

Note 6-5: The equation is listed as following. Please don't exceed the above recommended value.

$$\begin{aligned} \text{1st Lane N \& 2nd Lane N skew} &< 200\text{ns} \\ F_{\text{clk}} (\text{Typ.}) &= F_v (\text{Typ.}) \times T_h (\text{Typ.}) \times T_v (\text{Typ.}) \\ F_{\text{clk}} (\text{Min.}) &\leq F_v \times T_h \times T_v \leq F_{\text{clk}} (\text{Max.}) \end{aligned}$$

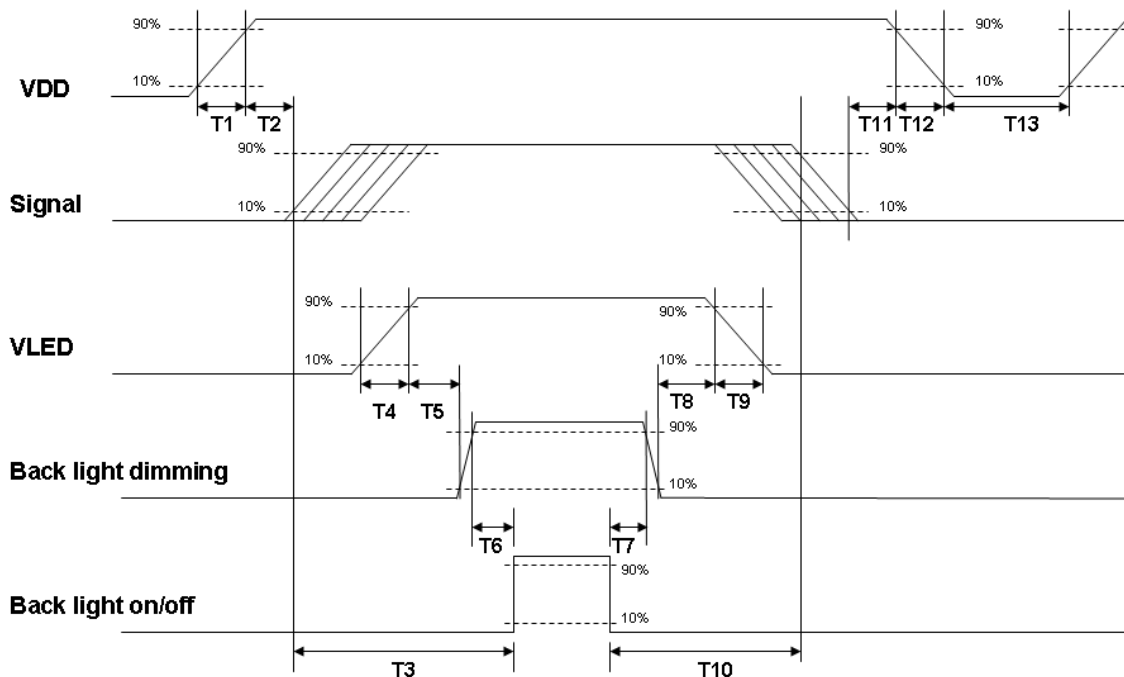
Before adjusting Fv, Th, and Tv, Fclk needs to meet recommended value first.

Note 6-6: The equation is listed as following. Please don't exceed the above recommended value.

$$F_v = F_{\text{clk}} (\text{Typ.}) / (T_v \times T_h)$$

6.7 Power ON/OFF Sequence

VDD power and lamp on/off sequence is as below. Interface signals are also shown in the chart. Signals from any system shall be Hi-Z state or low level when VDD is off.



Power sequence timing

Parameter	Value			Units
	Min.	Typ.	Max.	
T1	0.5	-	10	ms
T2	40	-	-*1	
T3	200	-	-	
T4	0.5	-	10	
T5	10	-	-	
T6	10	-	-*2	
T7	0	-	-	
T8	10	-	-	
T9	-	-	10	
T10	110	-	-	
T11	0	16	50	
T12	0	-	10	
T13	1000	-	-	

Note 1 (T2) : The maximum timing of VDD rising(90%) to HTPDN falling edge decided by customer system.

Note 2 (T6) : voltage of VDD must decay smoothly after power-off.(customer system decide this value)

7. Reliability Test Criteria

All testing under following setting: VDD 12V, Frame rate: 60Hz

Items	Required Condition	Note
Temperature Humidity Bias (Ta)	50°C/80%,300Hr	
High Temperature Operation (Ta)	50°C, 300Hr	
Low Temperature Operation (Ta)	0°C, 300Hr	
Hot Storage	60°C, 300 hours	
Cold Storage	-20°C, 300 hours	
Thermal Shock Test (Ta)	-20°C /30 min ,60 °C /30 min ,100cycles, 40°C minimum ramp rate	
Shock Test (Non-Operating)	50G,20ms,Half-sine wave,(+X,+Y,+Z)	
Vibration Test (Non-Operating)	1.0G, 10~300Hz, Random wave 10mins/axis, 3 direction (X, Y, Z)	
Altitude Test	Operation : 10,000ft Non-Operation : 30,000ft	
ESD	Contact : ± 8KV/ operation, Class B Air : ± 15KV / operation, Class B	Note 1

Note1: According to EN61000-4-2, ESD class B: Some performance degradation allowed. No data lost

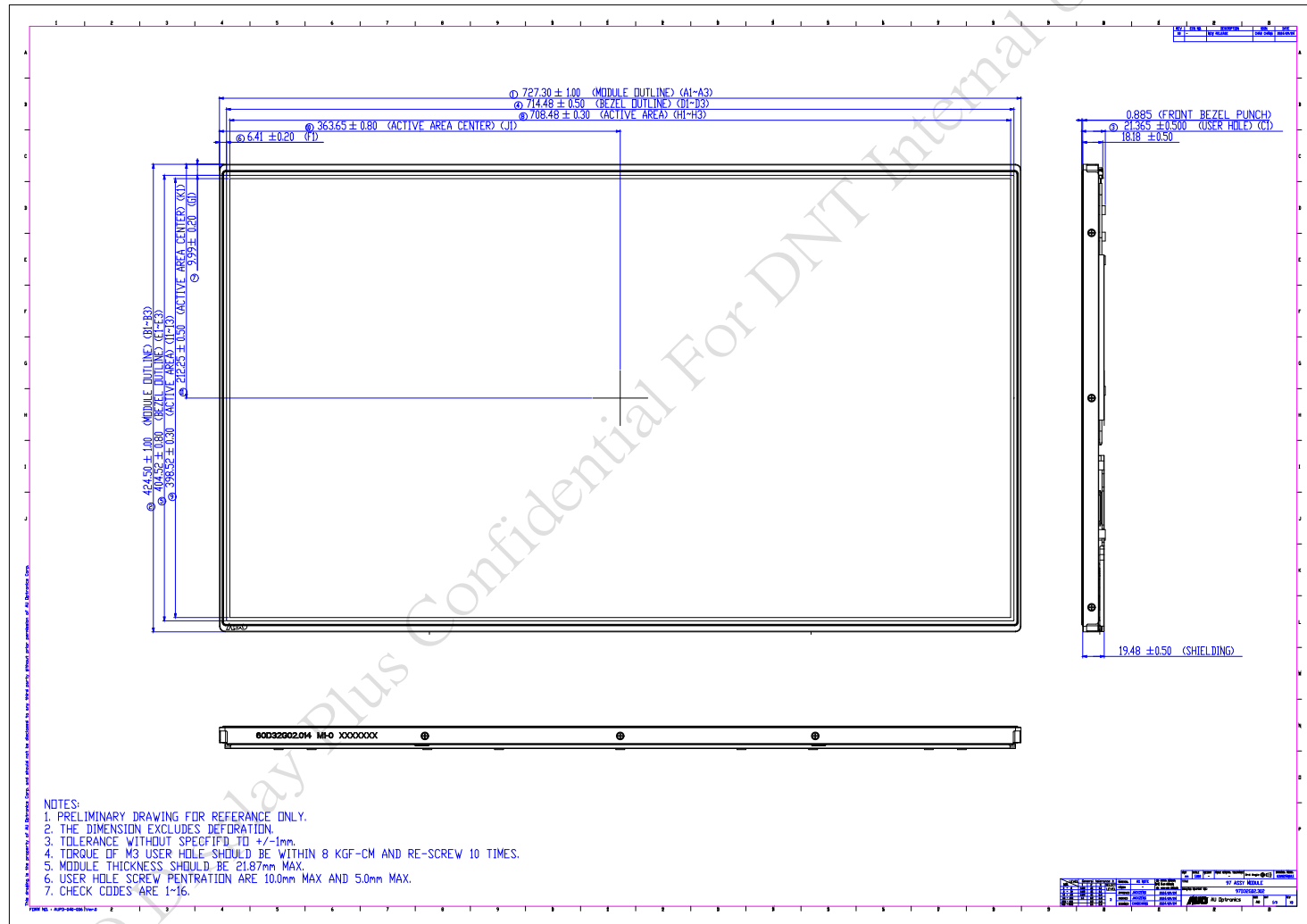
. Self-recoverable. No hardware failures.

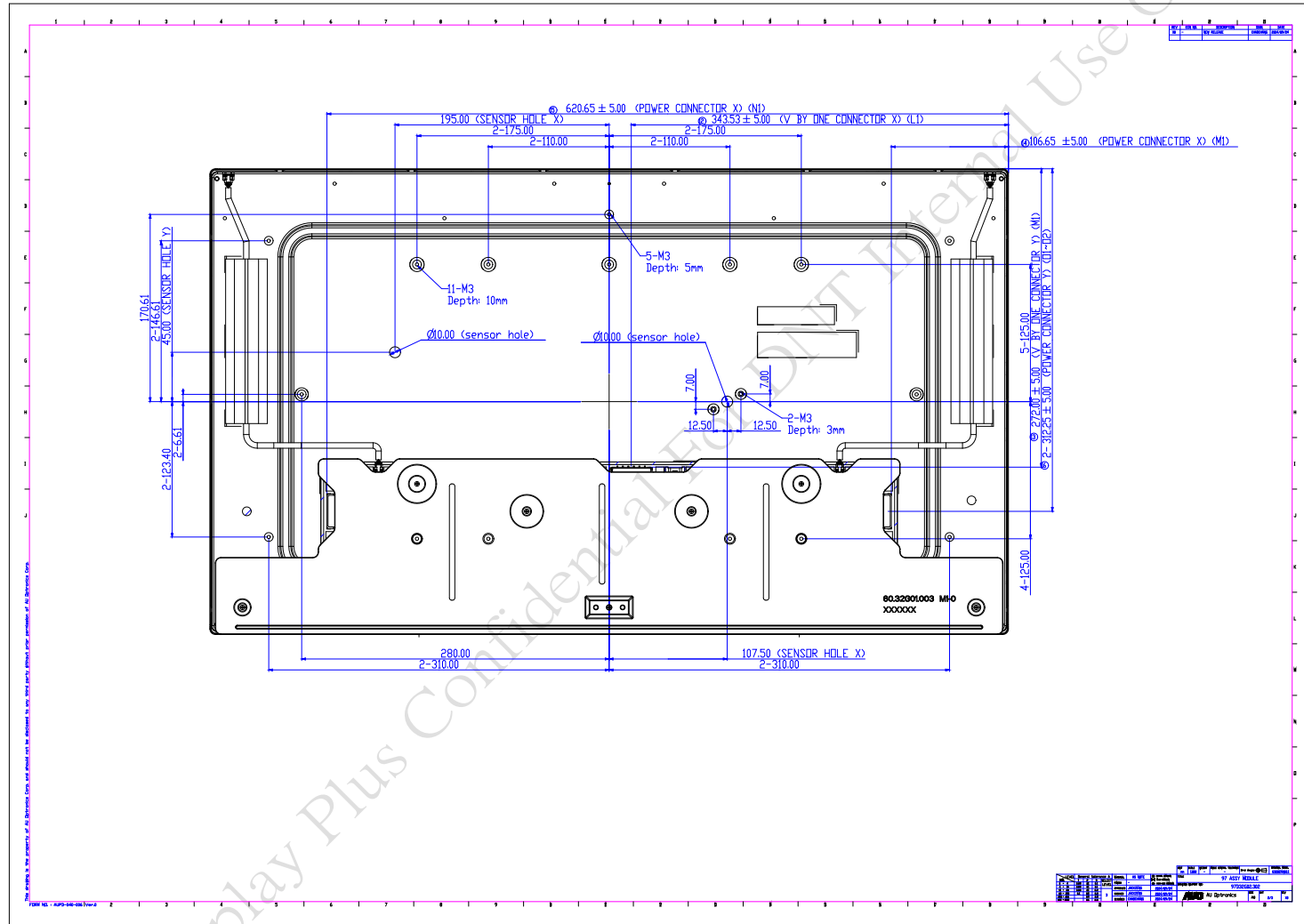
Note2:

- Water condensation is not allowed for each test items.
- Each test is done by new TFT-LCD module. Don't use the same TFT-LCD module repeatedly for reliability test.
- The reliability test is performed only to examine the TFT-LCD module capability.
- To inspect TFT-LCD module after reliability test, please store it at room temperature and room humidity for 24 hours at least in advance.
- No function failure occurs.

8. Mechanical Characteristics

8.1 LCM Outline Dimension



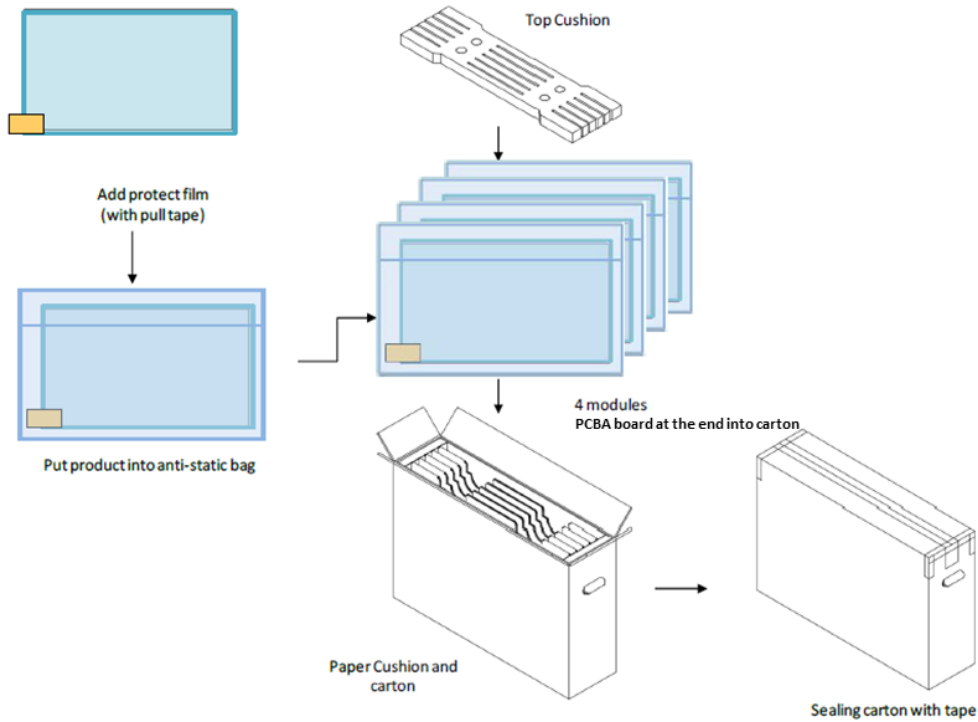


9. Label and Packaging

9.1 Shipping Label (on the rear side of TFT-LCD display)



9.2 Carton Package



Capacity: 4 modules /carton

Weight: 21.5±0.5 kg

Carton Dim.: 820(L)mm* 220(W)mm* 535(H)mm

Pallet Dim.: 1150(L)mm* 840(W)mm* 132(H)mm

10 Safety

10.1 Sharp Edge Requirements

There will be no sharp edges or comers on the display assembly that could cause injury.

10.2 Materials

10.2.1 Toxicity

There will be no carcinogenic materials used anywhere in the display module. If toxic materials are used, they will be reviewed and approved by the responsible AUO toxicologist.

10.2.2 Flammability

The printed circuit board will be made from material rated 94-V1 or better. The actual UL flammability rating will be printed on the printed circuit board.

10.3 Capacitors

If any polarized capacitors are used in the display assembly, provisions will be made to keep them from being inserted backwards.

10.4 International Safety Standard Compliance

The display module will satisfy all requirements for compliance to: IEC/UL62368-1